MILITARY STANDARD

MIL-STD-1320-73A
(NAVY)
12 NOVEMBER 1993

SUPERSEDING
MIL-STD-1320-73
16 SEPTEMBER 1993

TRUCKLOADING

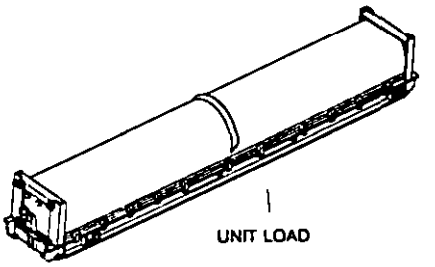
SPARROW III MISSILE, (RIM-7D, RIM-7E, RIM-7F, RIM-7M, AND RIM-7P) IN CONTAINER MK 470 MOD 0 AND MOD 1

CONTAINER DATA:

DIMENSIONS.....L-156, W-21, H-21
GROSS WEIGHT.....SEE TABLE I
TARE WEIGHT.....358 LBS
CUBE.....39.8 CU FT

△ DO NOT USE FOR SHIPPING WEIGHT

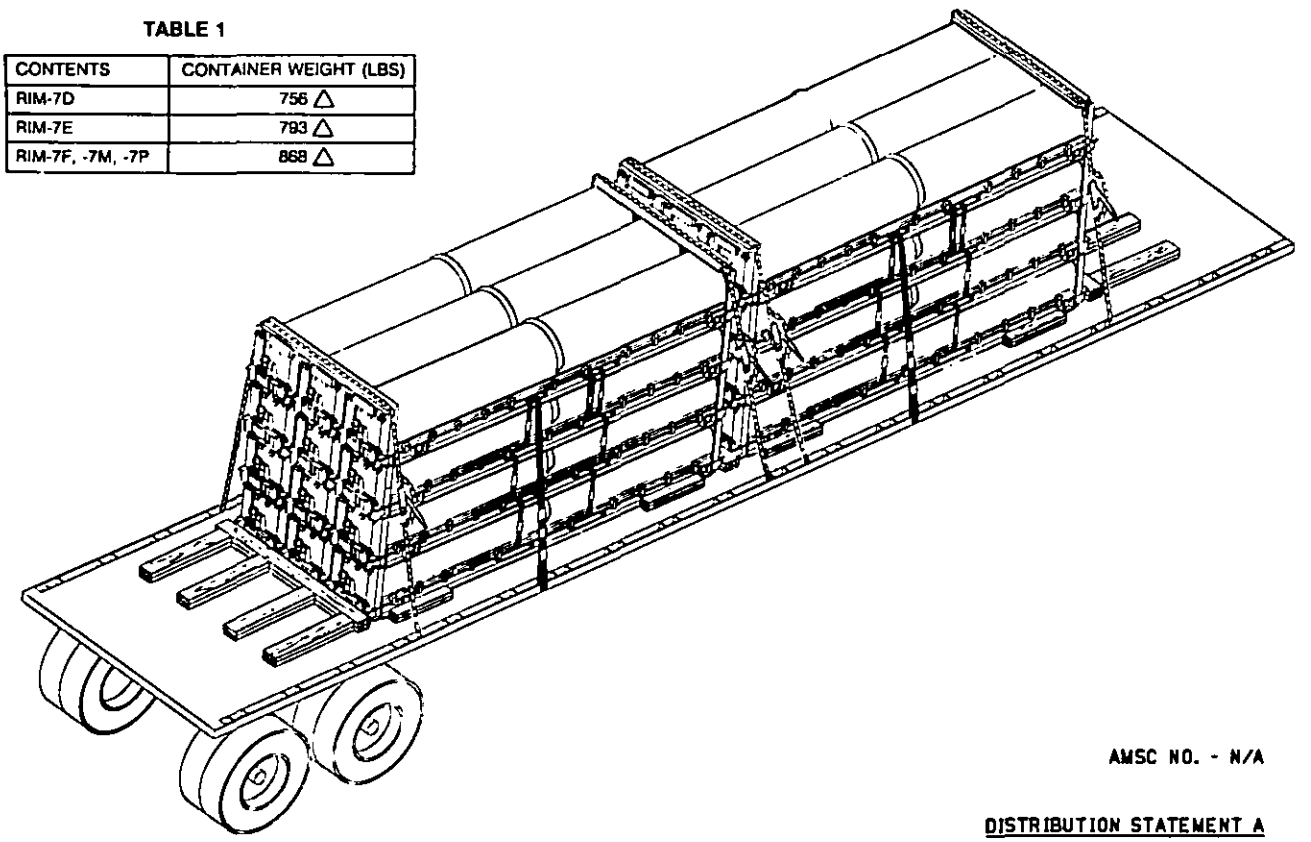
FOR HAZARD CLASSIFICATION
SEE SW020-AC-SAF-010/020/030



UNIT LOAD

TABLE 1

CONTENTS	CONTAINER WEIGHT (LBS)
RIM-7D	756 △
RIM-7E	793 △
RIM-7F, -7M, -7P	868 △



AMSC NO. - N/A

DISTRIBUTION STATEMENT A
APPROVED FOR PUBLIC RELEASE: DISTRIBUTION IS UNLIMITED.

NOTES:
1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, CONTAINERLOADING AND CARLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

REV LTR	REVISION DESCRIPTION	DATE	TDA	SYSCOM
A	ADDED CAUTION NOTE ON PG 7	1/2/93	CR	AE
			APPROVAL	

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENT ONLY.

SIGNATURE TDA, WPNSTA EARLE DATE 9/16/91

SIGNATURE AIRSYSCOM BY DIRECTION DATE 9/17/91

ORIGINATOR 9/12/91 DATE

WPNSTA EARLE, NEW JERSEY

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GENERAL NOTES:

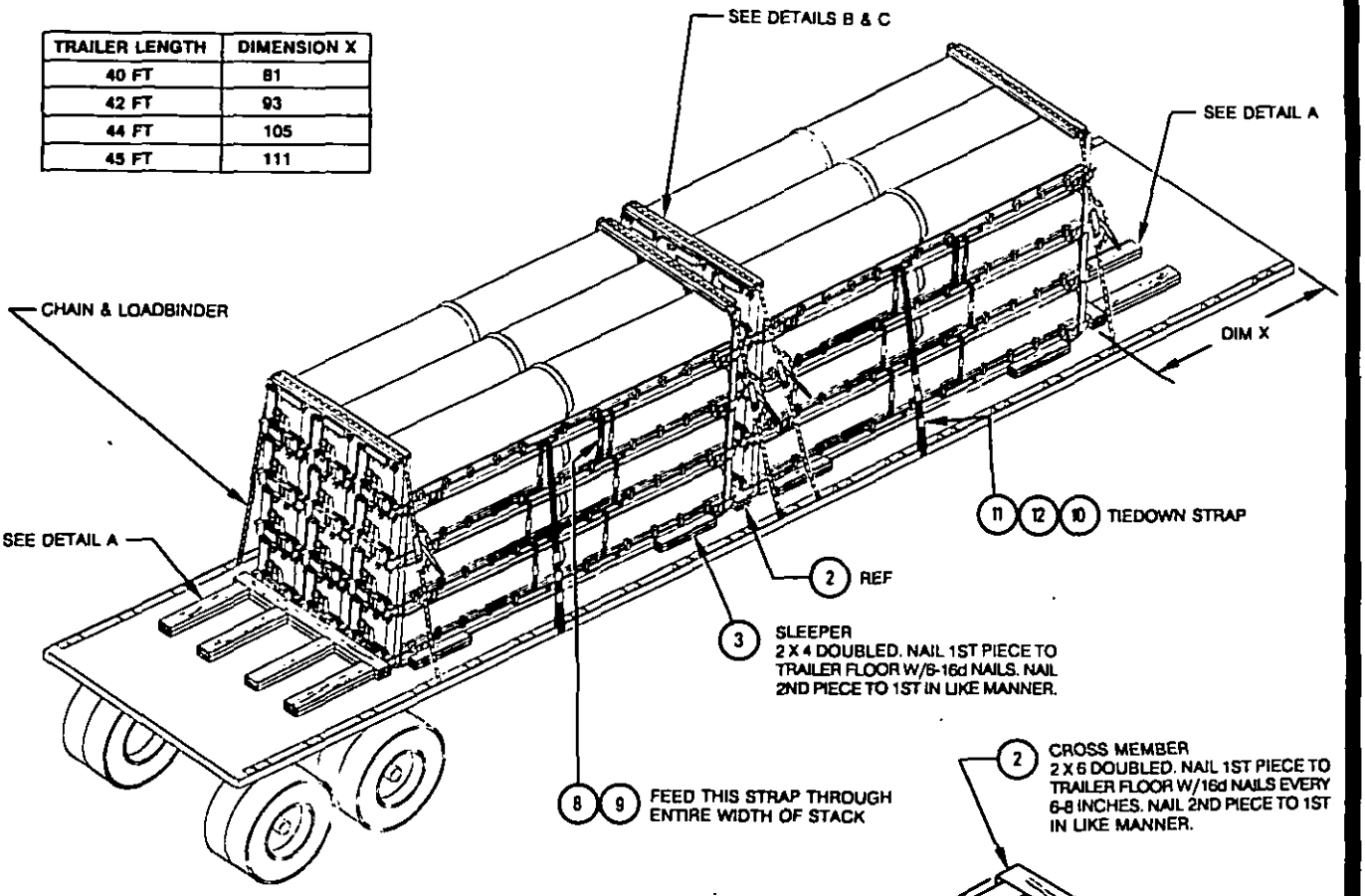
1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE SPARROW III MISSILE RIM-7D, RIM-7E, RIM-7F, RIM-7M, AND RIM-7P IN THE SHIPPING AND STORAGE CONTAINERS MK 470 MOD 0 AND MOD 1.
2. *THE PROCEDURES DESCRIBED HEREIN ARE INTENDED FOR 40 THRU 48 FT FLATBED TRAILERS WITH THE TRAILER'S TANDEM* AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH ALL METAL FLOORS.
3. CHAINS AND STEEL STRAPPING MAY BE USED INTERCHANGEABLY ON A ONE-TO-ONE BASIS FOR TIEDOWNS. CHAINS MAY NOT BE SUBSTITUTED FOR THE CENTER TIEDOWN STRAP, ITEM 10.
4. STEEL STRAPPING USED AS TIEDOWNS SHALL BE 2 X .050 AND SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).
5. CHAINS, FITTINGS, AND LOAD BINDERS SHALL CONFORM TO AND BE APPLIED AS SPECIFIED IN THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY), EXCEPT THAT THE CHAIN/GRAB HOOKS SHALL BE ATTACHED TO THE STAKE POCKETS (NOT AROUND THE RUB RAIL).
6. THE PROTECTOR BOARD ASSEMBLY SHOWN IN DETAIL B ON PAGE 3 SHALL BE USED UNDER THE END TIEDOWNS. STRAPS SHALL BE SECURED TO THE PROTECTOR BOARDS USING 2-INCH STRAPPING STAPLES. WHEN USING CHAIN, THE CHAINS SHALL BE SECURED TO THE PROTECTOR BOARDS BY DRIVING A 10d NAIL THROUGH THE CHAIN LINK AND CLINCHING THE NAIL OVER THE CHAIN. FIVE STAPLES/NAILS SHALL BE USED FOR EACH TIEDOWN.
7. BEFORE LOADING ONTO THE TRAILER, CONTAINERS SHALL BE UNITIZED AS SHOWN ON PAGE 7. CONTAINERS MAY BE UNITIZED FOUR HIGH.
8. THE FTL PLAN ON PAGE 3 SHOWS EACH STACK BEING 3 CONTAINERS WIDE. STACKS MAY BE FOUR CONTAINERS WIDE PROVIDED THERE IS SUFFICIENT NAILING SPACE ON THE TRAILER FLOOR FOR THE SIDE BLOCKING, ITEM 3. STACKS MAY ALSO BE FOUR WIDE WHEN LOADING ONTO 102" WIDE TRAILERS.
9. PRIOR TO INSPECTION REQUIRED BY NAVSEA OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS, AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR, AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS, FITTINGS, AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 22 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
10. *UNLESS OTHERWISE SPECIFIED, DUNNAGE LUMBER, STEEL STRAPPING, STRAP SEALS, AND NAILS SHALL BE AS SPECIFIED* IN THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY).
11. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS, AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
12. AFTER THE BLOCKING, BRACING, AND TIEDOWNS HAVE BEEN INSPECTED, THE CONTAINERS SHALL BE COMPLETELY COVERED WITH FIRE RESISTANT, WATERPROOF TARPAULINS. THE TARPAULINS MAY BE UNDER THE TIEDOWNS AND THE PROTECTOR BOARDS.
13. AFTER THE TARPAULINS ARE IN PLACE, ATTACH THE PROPER EXPLOSIVES PLACARDS TO BOTH SIDES, FRONT, AND BACK OF THE TRAILER, AND ATTACH THE SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA ON THE BACK DECK OF THE TRAILER.
14. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320 (NAVY).

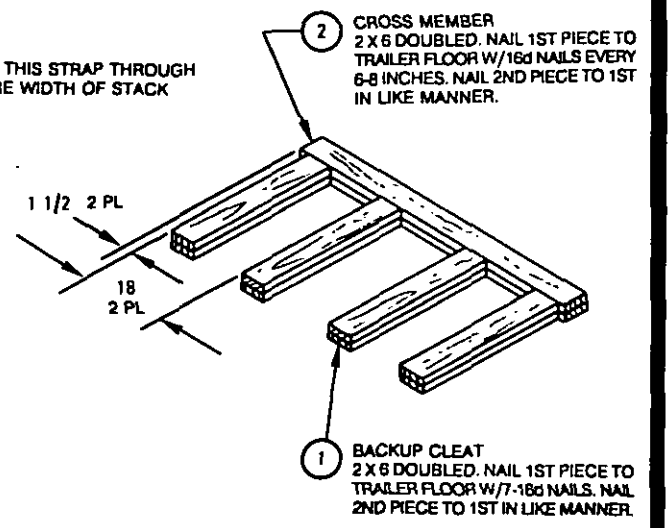
LIST OF MATERIALS

REQ'D	ITEM	DESCRIPTION	MAT'L/DWG	DIMENSIONS
AS REQ	16	CROSS MEMBER (LTL)	WOOD	2 X 6 X 45
AS REQ	15	CROSS MEMBER (LTL)	WOOD	2 X 6 X 24
AS REQ	14	SPACER BLOCKING (LTL)	WOOD	2 X 6 X 20 1/2
4	13	CHAIN & LOADBINDER	STEEL	
12	12	SEAL	STEEL	2" SIZE
2	11	STAKE POCKET PAD	STEEL	2 X .050 X 18
2	10	TIEDOWN STRAP	STEEL	2 X .050 X 25 FT
38	9	SEAL	STEEL	1 1/4" SIZE
2	8	HORIZONTAL STRAP	STEEL	1 1/4 X .035 X 16 FT
36	7	UNITIZING STRAP	STEEL	1 1/4 X .035 X 9 FT
12	6	END BLOCK	WOOD	2 X 4 X 10
12	5	HORIZONTAL	WOOD	2 X 4 X 20 1/4
4	4	PROTECTOR BOARD	WOOD	2 X 6 X 62 1/4
16	3	SLEEPER	WOOD	2 X 4 X 24
6	2	CROSSMEMBER	WOOD	2 X 6 X 66
16	1	BACKUP CLEAT	WOOD	2 X 6 X 36

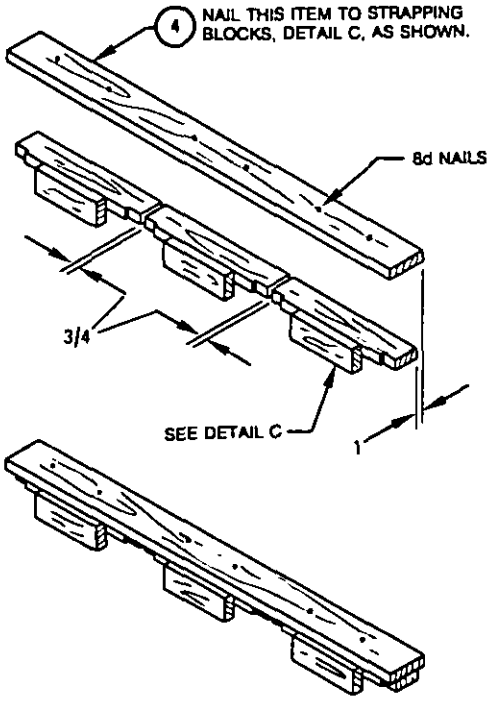
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TRAILER LENGTH	DIMENSION X
40 FT	81
42 FT	93
44 FT	105
45 FT	111

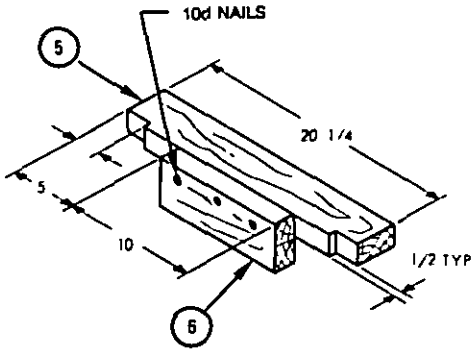




DETAIL A



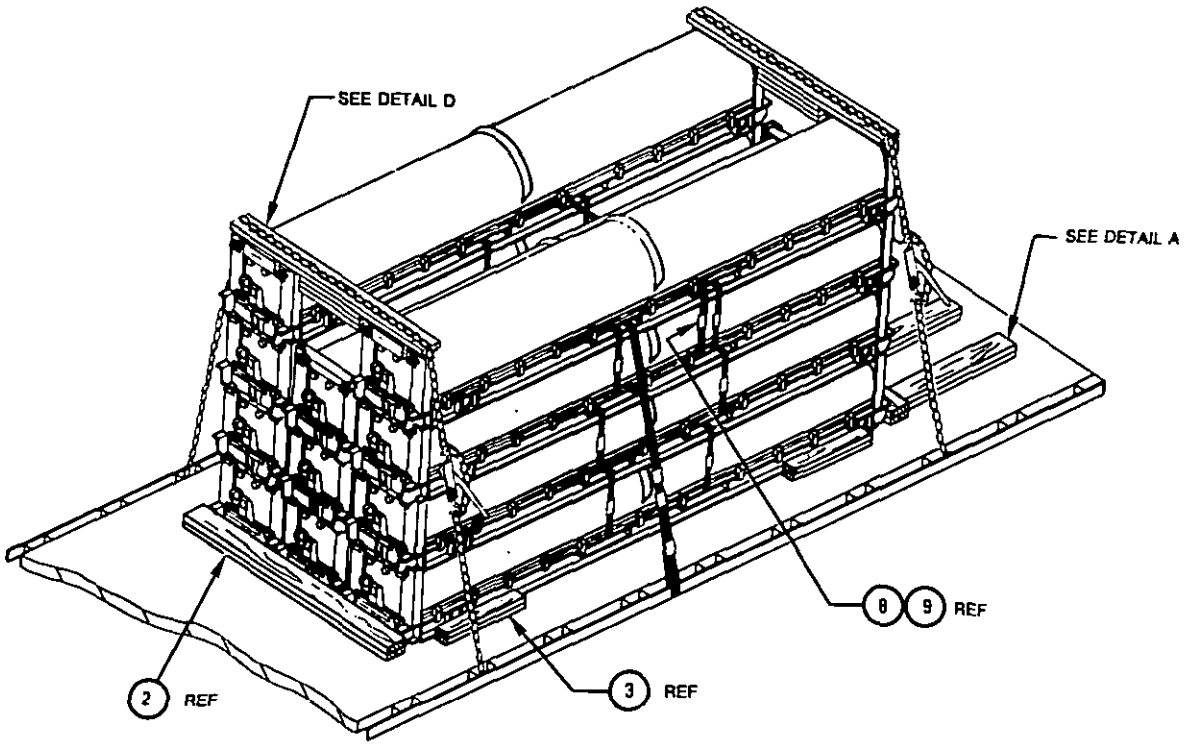
DETAIL B



DETAIL C
STRAPPING BLOCK

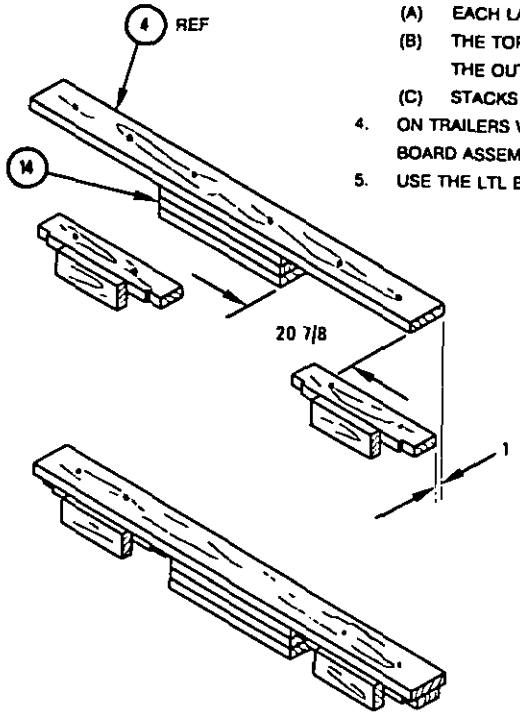
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LTL 40 THRU 48 FT FLATBED TRAILER

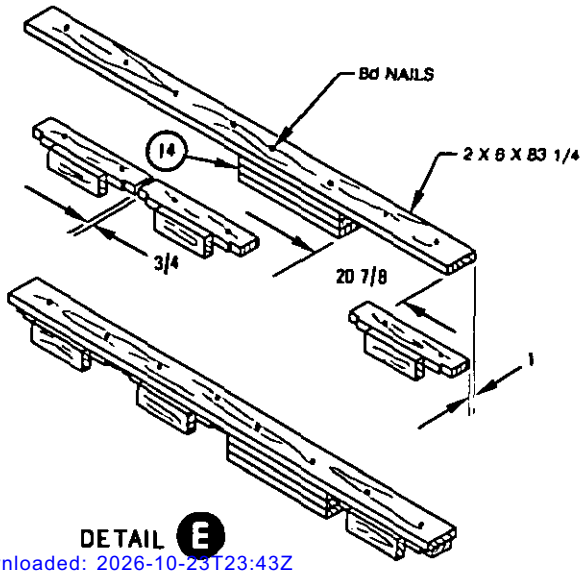


LTL NOTES:

1. THE ILLUSTRATION ABOVE SHOWS THE USE OF DETAIL D WHEN THERE ARE ONLY TWO CONTAINERS IN THE TOP LAYER OF A STACK.
2. THE DIAGRAMS ON THE FOLLOWING PAGE SHOW THE SUGGESTED LOAD PATTERNS FOR THE NUMBERS OF CONTAINERS TO BE SHIPPED.
3. LTL SHIPMENTS CAN BE ARRANGED ON THE TRAILER IN ANY MANNER UNDER THE FOLLOWING CONDITIONS:
   (A) EACH LAYER OF A STACK MUST HAVE THREE CONTAINERS.
   (B) THE TOP LAYER OF A STACK MAY HAVE ONLY 2 CONTAINERS IF THE TWO CONTAINERS ARE PLACED ON THE OUTSIDE ROWS AS SHOWN IN THE ILLUSTRATION ABOVE.
   (C) STACKS WHICH ARE THREE OR FOUR CONTAINERS HIGH CANNOT BE LESS THAN THREE ROWS WIDE.
4. ON TRAILERS WHICH PERMIT FOUR-WIDE STACKS (SEE GENERAL NOTE 8), USE DETAIL E AS THE PROTECTOR BOARD ASSEMBLY WHEN THE TOP LAYER HAS ONLY THREE CONTAINERS.
5. USE THE LTL BLOCKING AND BRACING PRINCIPLES SHOWN IN DETAIL F ON PAGE 8.



DETAIL D



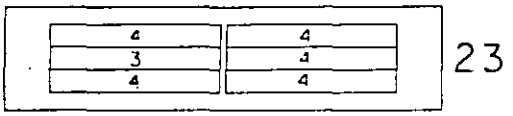
DETAIL E

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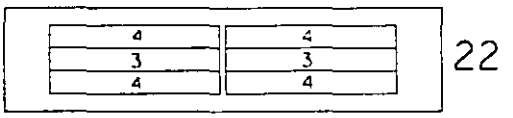
SUGGESTED LTL LAYOUT PLANS

FWD END →

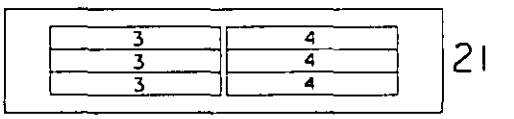
NUMBER OF CONTAINERS



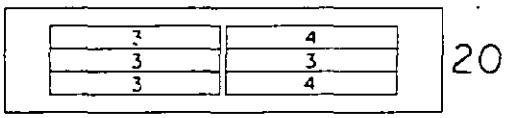
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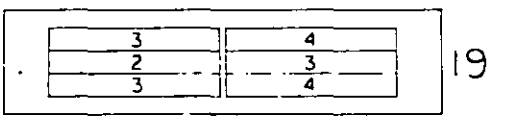
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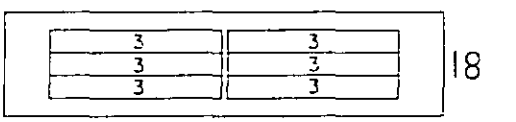
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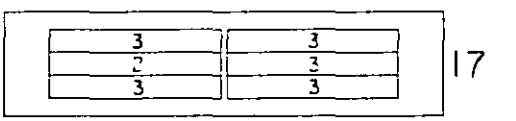
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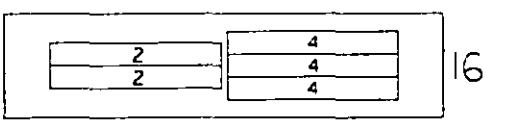
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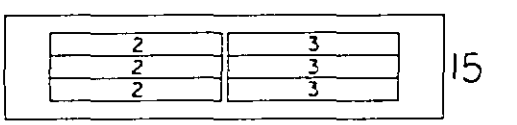
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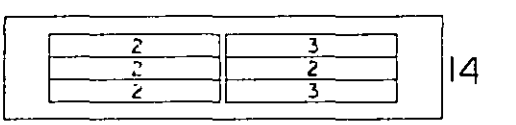
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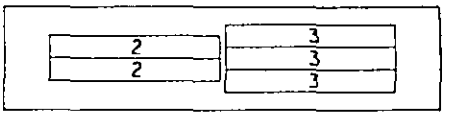
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14

FWD END →

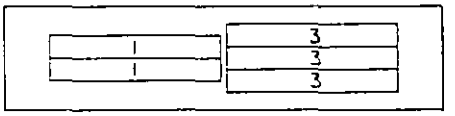
NUMBER OF CONTAINERS



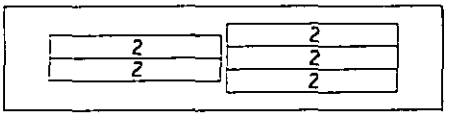
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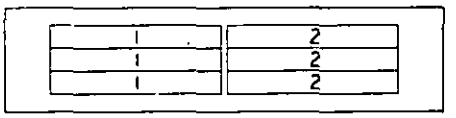
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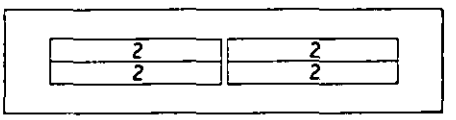
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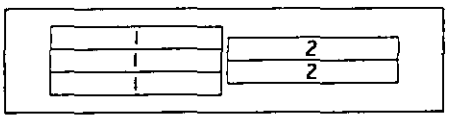
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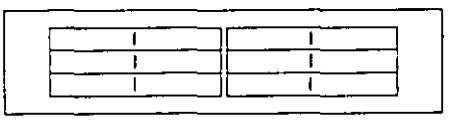
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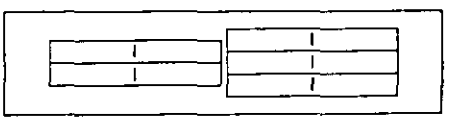
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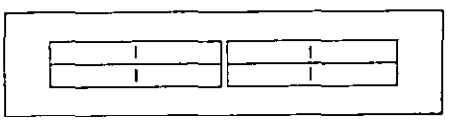
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6



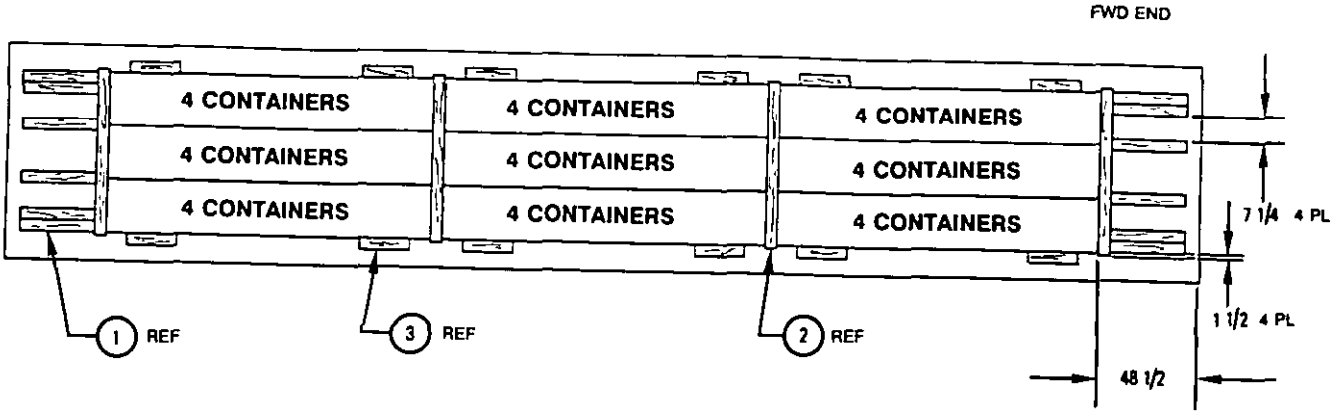
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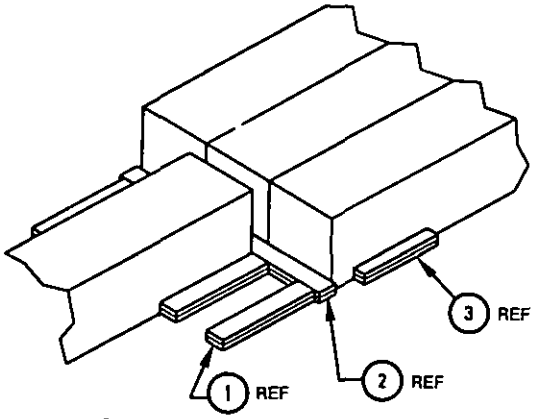


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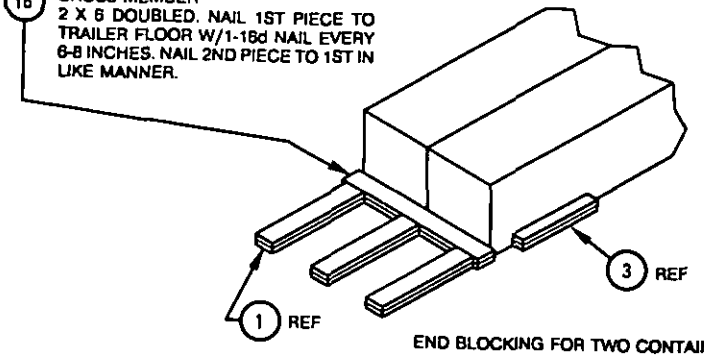
FTL 48 FT FLATBED TRAILER



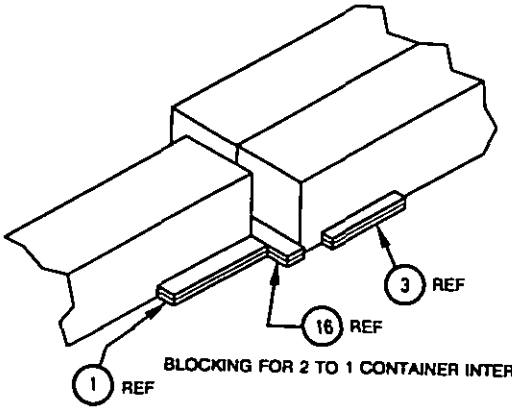


BLOCKING FOR 3 TO 1 CONTAINER INTERFACE

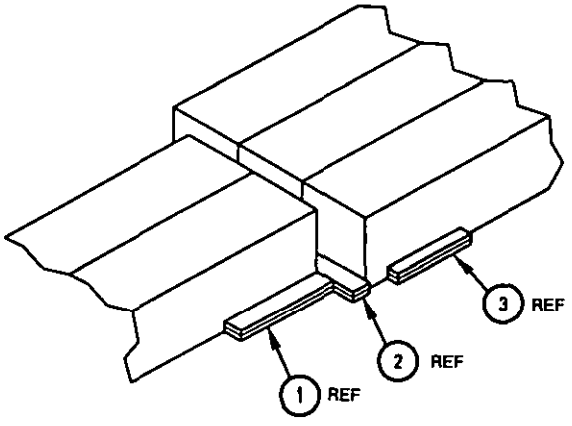
16 CROSS MEMBER
2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 6-8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.



END BLOCKING FOR TWO CONTAINERS

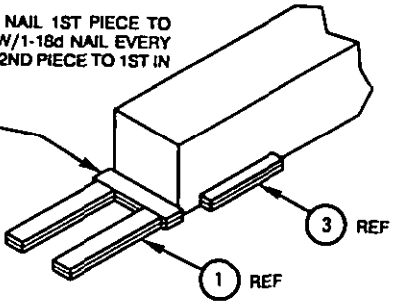


BLOCKING FOR 2 TO 1 CONTAINER INTERFACE



BLOCKING FOR 3 TO 2 CONTAINER INTERFACE

15 CROSS MEMBER
2 X 6 DOUBLED. NAIL 1ST PIECE TO TRAILER FLOOR W/1-16d NAIL EVERY 6-8 INCHES. NAIL 2ND PIECE TO 1ST IN LIKE MANNER.

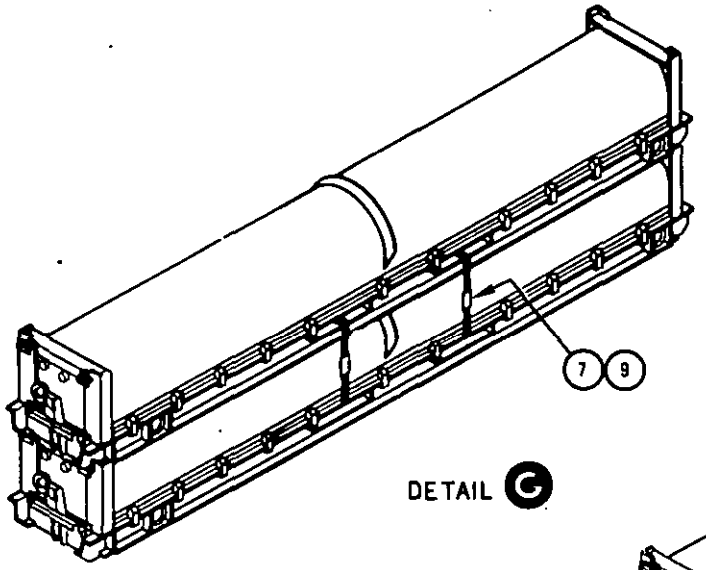


END BLOCKING FOR ONE CONTAINER

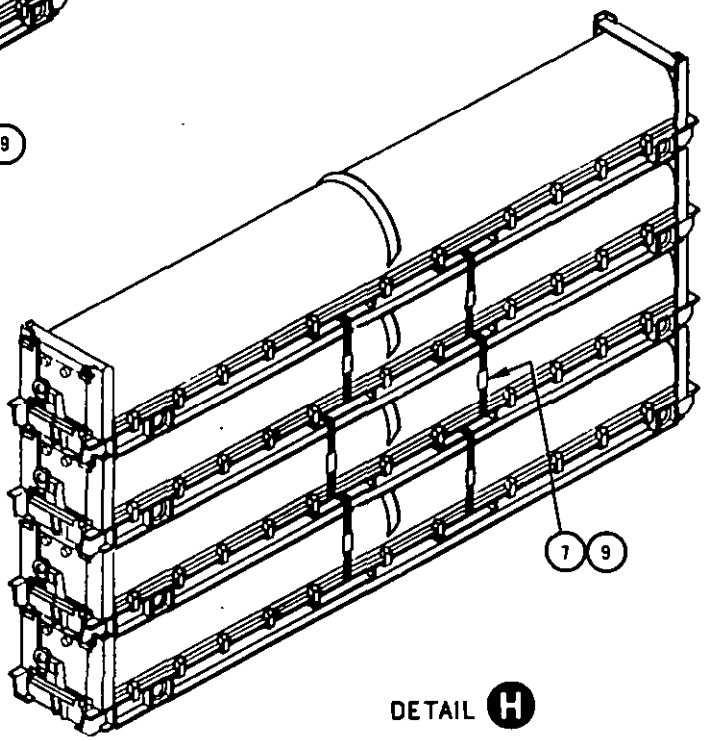
DETAIL F

BLOCKING AND BRACING FOR VARIOUS LTL COMBINATIONS

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DETAIL G



DETAIL H

WARNING

WHEN LOADING/UNLOADING TRAILERS WITH UNITIZED LOADS, EXTRA CAUTION SHOULD BE TAKEN TO PREVENT TOPPLING. SPECIAL ATTENTION SHOULD BE GIVEN TO APPROPRIATE BACKUP OF OUTBOARD UNITIZED LOADS.

UNITIZING PROCEDURE: (SEE CAUTION NOTE)

1. USING A FORK TRUCK, STACK ONE CONTAINER ON TOP OF ANOTHER AS SHOWN IN DETAIL G. BE SURE TO ALIGN THE STACKING FEATURES.
2. THREAD STRAPPING, ITEM 7, THROUGH THE FORK POCKETS OF BOTH CONTAINERS AS SHOWN IN DETAIL G. TENSION AND DOUBLE NOTCH THE SEALS (ONE SEAL PER STRAP).
3. REPEAT STEPS 1 AND 2 FOR TWO MORE CONTAINERS.
4. USING A FORK TRUCK, STACK ONE 2-CONTAINER UNIT LOAD ON TOP OF ANOTHER AS SHOWN IN DETAIL H. BE SURE TO ALIGN THE STACKING FEATURES.
5. THREAD STRAPPING, ITEM 7, THROUGH THE FORK POCKETS OF THE 2ND AND 3RD CONTAINERS AS SHOWN IN DETAIL H. TENSION AND DOUBLE NOTCH THE SEALS (ONE SEAL PER STRAP).

DEUNITIZING PROCEDURE: (SEE CAUTION NOTE)

1. CUT AND REMOVE STRAPS WHICH PASS THROUGH THE FORK POCKETS OF THE 2ND AND 3RD CONTAINERS.
2. REMOVE THE UPPER 2-CONTAINER UNIT LOAD.
3. CUT AND REMOVE ALL REMAINING STRAPS FROM 2-CONTAINER UNIT LOADS.
4. REMOVE TOP CONTAINERS.

CAUTION

THE BANDING TOGETHER OF THE CONTAINERS TO FORM A UNIT LOAD ARE FOR LOADING AND UNLOADING THE TRAILER ONLY. CONTAINERS ARE NOT TO BE TRANSFERED AT SEA OR HANDLED BANDED TOGETHER. WHEN LOADING/UNLOADING TRAILERS WITH UNITIZED LOADS, EXTRA CAUTION SHOULD BE TAKEN TO PREVENT TOPPLING. SPECIAL ATTENTION SHOULD BE GIVEN TO APPROPIATE BACKUP OF OUTBOARD UNITIZED LOADS.

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